🧊 3D Coordinates Extension 2 — English Worksheet (Advanced)

Topic: 3D Mob Heads (x, y, z) · **Course:** 3D Coordinates · **Level:** Advanced (Extension, after Lesson 3) · **Time:** about 45 minutes

Build **two mob heads** in 3D — a **creeper** and a **piglin**. Every block needs **three** numbers **(x, y, z)**: x **across**, y **up**, z **deeper** (forward).

> 🔴 Stand on your **home spot** (red block) every run. Move your feet, move your head.

1 · Predict 🔮

Read each set. Write your prediction **and the reason**.

```
place green block at (3, 5, 1)
place green block at (3, 5, 2)
place green block at (3, 5, 3)
```

Does the line go across, up, or deeper? Why?

```
place green block at (4, 5, 1)
place green block at (4, 6, 1)
place green block at (4, 7, 1)
```

Across, up, or deeper? Why?

```
place green block at (3, 5, 1)
place green block at (4, 5, 1)
place green block at (3, 5, 2)
place green block at (4, 5, 2)
```

How many blocks, and what shape do they make? Why?

```
place green block at (3, 5, 1)
place black block at (3, 5, 1)
```

Same (x, y, z). Green, black, or both? Why?

2 · Spot the Bug 🐛

Fix each one. Then say why it was wrong.

Bug A — This block should sit one step **deeper**, at (3, 5, 2). It only has two numbers, so it lands on the front.

```
place green block at (3, 5)
```

Fixed code:

Why wrong? Why does your fix work? (2–3 sentences)

Bug B — The eye should be **up high and one step deep**: (4, 7, 2). The last two numbers are swapped.

```
place black block at (4, 2, 7)
```

Fixed code:

Why wrong? Why does your fix work? (2–3 sentences)

Bug C — The code runs with no error, but the creeper's eye ended up on the **side** of the head instead of the front. The front face is built at z = 1, but this eye was placed at z = 2 — one block too forward.

```
place green block at (4, 7, 1)
place black block at (4, 7, 2)
```

Fix it so the eye is on the front face, and explain what "one block too forward" means. (2–3 sentences)

3 · Show Your Work 📸 🎥

Build **both** heads on your home spot — a **creeper** and a **piglin** — then add one change of your own.

Plan your coordinates first. Pick a home block for each head so they don't overlap.

> **Creeper:** build the front face at z = 1 (green square + black eyes + black mouth), then repeat at z = 2 and z = 3. Plain green where the face was.
>
> **Piglin:** build the front face at z = 1 (pink square + a wide black mouth + nose), then repeat at z = 2 and z = 3. Plain pink where the face was.
>
> **Face on front only:** at z = 2 and z = 3, fill the face spots with the plain head colour.

Then a MODIFY challenge: change one thing your own way — a new colour, an extra part (ears, hat), or make a head bigger.

> 💡 Tip: the eyes, nose, and mouth only show on the **front**. The sides and back are plain.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.
>
> The most fun part was _____.
>
> Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.